The Kagome Lattice and FFGFT:

Four Structural Parallels

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Abstract

Pei, Chen, Castelnovo and Moessner (PRL 137, 106702, 31 August 2026, Open Access) study the singly doped infinite-U Hubbard model on the kagome lattice. They find resonating-valence-bond (RVB) polarons, disorder-free self-trapping with band mass increasing by six orders of magnitude, and a crossover to $\sqrt{3} \times \sqrt{3}$ order in the unpolarised limit, which they benchmark against the three-state Potts model. This document identifies four structural parallels to FFGFT and supports each with the wording of the source. The parallels share one algebraic root: the Frobenius automorphism $\phi : x \mapsto x^3$ of order 3 on finite fields. A separate section states what does *not* correspond. Status of all four parallels: **[S]** (structural correspondence established, physical identification open).

Status Markers

[B]	Algebraically proved.
[K]	Numerically confirmed.
[S]	Structurally plausible; physical identification open.
[Q]	Taken from the source (Pei et al.), quoted verbatim.
[ASSUMPTION]	Axiomatic postulate.

All verbatim quotations are from Pei et al. (2026), Open Access under CC-BY; section references follow the structure of that paper.

1 The Source: What Pei et al. Show

Model and question

The model is the infinite-U Hubbard model on the kagome lattice with a single hole:

> **[Q]**"To gain deeper insight into the role of hole kinetics in determining magnetism in highly frustrated doped Mott insulators, we consider the single-hole counter-Nagaoka problem on the kagome lattice, using magnetization as a tuning parameter." (Abstract)

The term *counter-Nagaoka* matters: Nagaoka's theorem predicts ferromagnetism from the kinetic energy of a hole on bipartite lattices. On the kagome lattice the kinetics is *frustrated*, and antiferromagnetic correlations appear instead:

> **[Q]**"When said kinetic energy is frustrated, it was later shown that antiferromagnetic correlations can also appear." (Introduction)

The physics of a single triangle

The mechanism is visible on a single triangle. Pei et al. describe it:

> **[Q]**"Ferromagnetic spins have lowest energy $E_0 = -t$ due to destructive interference of the hole motion around the triangle; whereas antiferromagnetic spins can form a singlet and effectively thread a π flux through the triangle, relieving the frustration and allowing the hole to delocalize and lower the energy to $E_0 = -2t$ (bottom of an unfrustrated 1D tight-binding band)." (Section *Model*)

Two points to retain. First: the ferromagnetic configuration is *destructively interfered* — the hole cannot move freely. Second: the singlet *threads a π flux* through the triangle. This π flux is not a metaphor but a Berry phase: the hole accumulates phase π on one full circuit, and the interference switches from destructive to constructive.

Flat band and compact localised states

In the fully polarised case the problem reduces to spinless fermions. The lowest band is then exactly flat:

> **[Q]**"The fully polarized case reduces to a well-known spinless fermionic tight-binding model, where the bottom band is exactly flat (energy -2) and the states are compact localized around individual hexagons." (Section *RVB polarons and delocalization*)

Kinetic frustration arises precisely when this flat band is lowest:

> **[Q]**"Indeed, kinetic frustration arises when the kagome flat band happens to be lowest in energy for a hole-doped fully polarized electron system close to half-filling. As such, holes in a fully polarized system occupy compact localized states at low energy." (Introduction)

RVB polarons: formation, delocalisation, self-trapping

Adding a few reversed spins ($n = 1, \ldots, 4$) forms polarons that *delocalise* — but with strongly enhanced band mass:

> **[Q]**"Upon adding few reversed spins, polarons form and delocalize, albeit with a band mass 100 times larger than that of an unfrustrated tight-binding particle on the same lattice." (Introduction)

The structure of these polarons is described by valence-bond singlets on tree-like Husimi cacti:

> **[Q]**"The short distance physics of the kagome lattice is close to tree-like, and we find that the valence bond language of the same Hamiltonian on the Husimi cactus proves highly useful to model and understand the polarons in this Letter." (Section *RVB polarons and delocalization*)

At $n = 5$ the behaviour changes abruptly:

> **[Q]**"The polaron bandwidth suddenly drops by at least six orders of magnitude ($< 10^{-8}$ in a 4×4 system of 48 sites, with 16 near-degenerate GSs); to the best of our numerics it is indistinguishable from a flat band of localized states." (Section *Strong self-trapping*)

The mechanism of this self-trapping is a resonance of six symmetry-related clusters around a hexagonal plaquette:

> **[Q]**"[...] there are six symmetry-related such clusters around any hexagonal plaquette of the lattice, and the corresponding six Husimi cactus states are not orthogonal. [...] the GS wave function takes the form of a symmetric superposition of all six states, spreading across all 12 sites around the hexagonal plaquette." (Section *Strong self-trapping*)

And the interpretation:

> **[Q]**"By resonating around a hexagonal plaquette, it seems that the 1H5S RVB polaron acquires a possibly infinite mass, undergoing localization reminiscent to the compact-localized hexagonal states of the spinless single-particle problem." (Section *Strong self-trapping*)

Notably, this localisation occurs in a *disorder-free* system: the authors speak of states that "relocalize in a disorder-free system" (Conclusions).

The unpolarised limit and the Potts model

As n grows, the valence-bond language loses its usefulness and semiclassical correlations emerge:

> **[Q]**"We see that this 'frustration on frustration' leads to the appearance of magnetic correlations consistent with semiclassical $\sqrt{3}\times\sqrt{3}$ order as we approach the unpolarized limit." (Section *Unpolarized state*)

The benchmark is the three-state Potts model:

> **[Q]**"The data presented in Fig. 5 include our Letter of the three-state Potts model, with Hamiltonian $H_{\text{Potts}} = \sum_{\langle ij\rangle} \delta_{\sigma_i\sigma_j}$, $\sigma_i = 0, 1, 2$. The ground state ensemble is extensively degenerate and consists of all states without identical nearest neighbors." (Appendix *Potts model*)

This model is not chosen arbitrarily — it marks a phase transition:

> **[Q]**"This model is useful in that it sits at the Kosterlitz-Thouless (KT) transition separating a $\sqrt{3}\times\sqrt{3}$ ordered and an algebraically correlated phase, and it can thus provide a natural benchmark for an analysis of semiclassical correlations." (Section *Unpolarized state*)

2 Parallel 1: Frobenius Orbits — the Same Algebraic Structure

FFGFT side

[B] (Doc. 339): The multiplicative group $GF(27)^* \cong \mathbb{Z}_{26}$ decomposes under the Frobenius automorphism $\phi : x \mapsto x^3$ into

- two fixed points $\{+1, -1\}$ — the massive $\mathbb{Z}_2$ sector (particle/antiparticle),
- four three-element orbits in $\mathbb{Z}_{13}$, each $\{a, 3a, 9a \pmod{13}\}$: $\{1, 3, 9\}$, $\{2, 5, 6\}$, $\{4, 10, 12\}$, $\{7, 8, 11\}$,
- combined with the two $\mathbb{Z}_2$ parities: eight gluons of the adjoint $SU(3)$ representation.

The three-element orbits correspond to the three colour states $\{R, G, B\}$; the parity distinguishes colour from anticolour (Doc. 339, §3.2).

Kagome side

The kagome lattice has three sublattice sites per unit cell; the unit cell itself is a triangle. Pei et al. explicitly use system sizes that are multiples of 3:

> **[Q]**"$\sqrt{3}\times\sqrt{3}$ order requires L_x and L_y to be integer multiples of 3 to avoid incommensurability issues." (Appendix *Methods*)

The order itself shows at the K points of the Brillouin zone:

> **[Q]**"In all cases, we find peaks at the K points, suggestive of $\sqrt{3}\times\sqrt{3}$ order." (Section *Unpolarized state*)

The K points carry threefold rotational symmetry (C_3); the structure factor $S(\mathbf{k})$ is invariant under rotation by $2\pi/3$.

Structural correspondence

Both sides carry the same algebra: a cyclic group on which an automorphism of order 3 acts and generates three-element orbits. In FFGFT it is ϕ on $GF(27)^*$; on the kagome lattice it is C_3 on the sublattice sites and — in reciprocal space — on the K points. **[S]**

What is *not* claimed: that the four $\mathbb{Z}_{13}$ orbits are physically identical to four kagome configurations. The kagome lattice has a single C_3 orbit per unit cell; FFGFT has four orbits in $\mathbb{Z}_{13}$. The correspondence lies in the *order* of the acting automorphism, not in the number of orbits.

3 Parallel 2: Three-State Potts Model = $GF(3)$ = Fixed Field

FFGFT side

[B] (Doc. 339, §3.3): The fixed field of ϕ on $GF(27)$ is $GF(3) = \{0, +1, -1\}$ (Fermat's little theorem: $x^3 = x$ for all $x \in GF(3)$). In FFGFT, $GF(3)$ corresponds to the $U(1)$ sector: the projection $GF(27) \to GF(3)$ describes the electrically neutral sector (photon). The three elements $\{0, +1, -1\}$ are also the three $\mathbb{Z}_3$ sectors of Doc. 336 (fixed points $\{0, +1, -1\}$ = FFGFT sectors **[B]**).

[B] (Doc. 321, §1.2): The triality condition $T_R = 0 \pmod 3$ is equivalent to confinement:

> "Confinement is equivalent to requiring all observable states to have triality $T_R = 0$." (Doc. 321)

A single quark ($T_R = 1$) is not observable; a meson ($T_R = 1+2 = 0$) and a baryon ($T_R = 1+1+1 = 0$) are.

Kagome side

The Potts benchmark has exactly three states $\sigma \in \{0, 1, 2\}$ with $\mathbb{Z}_3$ symmetry (quoted above). The ground-state condition "no identical nearest neighbors" is a frustration condition: on a triangle all three states must occur — exactly one $\mathbb{Z}_3$-neutral configuration per triangle. The KT transition separates the phase in which this $\mathbb{Z}_3$ order is long-ranged ($\sqrt{3} \times \sqrt{3}$) from the algebraically correlated phase.

Structural correspondence

The identification $\sigma \in \{0, 1, 2\} \leftrightarrow GF(3) = \{0, +1, -1\}$ is algebraically exact: both are $\mathbb{Z}_3$ as an additive group. The Potts frustration condition (all three states on every triangle) and the FFGFT confinement condition (sum of trialities $\equiv 0 \pmod 3$) are the same statement: only $\mathbb{Z}_3$-neutral configurations are admissible. **[S]**

The Potts KT transition therefore has an FFGFT counterpart: the transition between confined ($T_R = 0$ enforced at long range) and deconfined (triality only algebraically correlated). Whether this identification holds physically beyond the algebra is open (§7).

4 Parallel 3: Mass as Frozen Kinetic Energy

FFGFT side

[B] (Doc. 286): The duality $\tilde{T} \cdot m = 1$ enforces two energy types:

	Static sector	Kinetic sector
Form	$E_0 = mc^2$ (rest mass)	$E = hf$ (free)
Seat	$\mathbb{Z}_3$ core (winding)	window
Time	$\tilde{T} = 1/m$	$\tilde{T} = 1/\omega$
Status	derived (Doc. 285)	open (Doc. 286)

Doc. 286 states: "The rest mass is static: a topologically stable winding number on T^4 [...] *Mass is frozen kinetic energy.*" **[B]** (Doc. 339): The massive/massless separation is algebraically forced by the Frobenius — fixed points $\{+1, -1\}$ are massive, the three-element orbits are massless (gluons).

Kagome side

The kagome system shows *the same dichotomy*, but not as a sector separation — as a *non-monotonic band mass* along the polarisation axis:

- $n = 0$: flat band, compact localised — "effectively infinite mass" (kinetic energy frozen by destructive interference).
- $n = 1, \ldots, 4$: delocalisation with band mass $\times 100$ ("polarons form and delocalize").
- $n = 5$: relocalisation, band mass $\times 10^6$ ("acquires a possibly infinite mass").
- $n \to$ unpolarised: delocalisation ("delocalized, dispersive holes are recovered").

Pei et al. summarise:

> **[Q]**"Our Letter uncovers an intricate interplay between single-particle flat bands and cooperative 'polaronic' self-trapping physics." (Introduction)

Structural correspondence

In both systems, *mass* results from geometric freezing of kinetic freedom: in FFGFT through topological winding on T^4 (Doc. 314: "winding numbers are topological"), on the kagome lattice through destructive interference around plaquettes. Both are localisation *without disorder* — a purely geometric effect. **[S]**

The difference: FFGFT has two *sectors* (massive/massless); the kagome system has one *parameter* (polarisation) along which band mass switches between the two extremes. The kagome system thus realises the transition that FFGFT treats as a sector boundary.

5 Parallel 4: π-Flux = Berry Phase = Winding Number

Kagome side

The quotation from §1.2 is central here: the singlet "effectively thread[s] a π flux through the triangle". The π flux is the phase the hole picks up on one circuit of the triangle. Without it (ferromagnetic) the hopping amplitude interferes destructively; with it, constructively. The energy gain is a factor 2 ($E_0 = -t \to -2t$).

The same logic repeats at larger scale in the self-trapping: the six Husimi states around the plaquette interfere *symmetrically* — constructive interference for the bound component, spreading the state over all twelve sites, but destructive for any motion *out* of the plaquette.

FFGFT side

[B] (Doc. 314, Ch. D2): "in the rolled-up state there are no fractal corrections — winding numbers are topological, the defect $d = 100\xi$ accumulates only when unrolled." Winding numbers on T^4 are quantised Berry phases around closed loops: phase $2\pi n$, $n \in \mathbb{Z}$, on one circuit of a torus cycle.

[B] (Doc. 321, §2.1): The $\mathbb{Z}_3$ generator τ acts as a unitary operator on $L^2(T^4)$ with $\tau^3 =$ id; eigenvalues are ω^k, $\omega = e^{2\pi i/3}$. The three sectors P_k are distinguished by the phase accumulated under a $\mathbb{Z}_3$ rotation.

[B] (Doc. 230): Entanglement is "holonomy around cycles" — the phase accumulated on a closed circuit is what constitutes entanglement geometrically.

Structural correspondence

The π flux in the kagome triangle (Berry phase π, half winding) and the $\mathbb{Z}_3$ phases ω^k in FFGFT (Berry phase $2\pi k/3$) are the same operation: phase accumulated along a closed path that changes the accessible Hilbert space. On the kagome lattice it lifts the frustration; in FFGFT it selects the sector. **[S]**

6 What Does Not Correspond

Honesty requires naming the limits of the parallels.

Dimension. The kagome lattice is two-dimensional; T^4/Z_3 is four-dimensional. The C_3 symmetry of the kagome lattice lives in the plane; the $\mathbb{Z}_3$ action on T^4 has nine fixed points (Doc. 314) and acts on a compact space without boundary. There is no map embedding the kagome lattice into T^4/Z_3.

Classical versus quantum. The Potts model is classical; Pei et al. use it as a *semiclassical* reference and scale correlators by $(S^z)^2 = 1/4$ instead of $S(S+1) = 3/4$. FFGFT works on $L^2(T^4) \otimes \mathbb{C}^3$ (Docs. 230/282) fully quantum-mechanically. The correspondence in Parallel 2 is algebraic ($\mathbb{Z}_3$), not dynamical.

Number of orbits. FFGFT has four three-element orbits in $\mathbb{Z}_{13}$ (eight gluons); the kagome lattice has one C_3 orbit per unit cell. Parallel 1 concerns the *order* of the automorphism, not the orbit count.

Structure versus dynamics. All four parallels are structural. None predicts that an FFGFT computation reproduces the kagome spectrum or vice versa. They state: the same algebra (ϕ of order 3) generates the same kind of constraint in both systems (only $\mathbb{Z}_3$-neutral configurations admissible; localisation without disorder; Berry phase as selector).

One unexplained number. The self-trapping cluster spans twelve sites around a hexagonal plaquette. In Doc. 314 the first excitation of the D_4 lattice has degeneracy $24 = 12 + 12$ ("FCC half and winding half separately addressable"). Whether the 12 of the kagome cluster relates to one of the 12-halves of D_4 is *not* investigated and is not listed here as a parallel. **[S]**

7 Summary and Open Questions

Kagome system (Pei et al.)	FFGFT	Shared algebra	Status
K-point order, C_3 on sublattice	Frobenius orbits on $GF(27)^*$ (Doc. 339)	automorphism of order 3	**[S]**
Three-state Potts, KT transition	$GF(3)$ fixed field, triality $T_R = 0$ (Docs. 339, 321)	$\mathbb{Z}_3$ neutrality	**[S]**
Flat band / self-trapping, band mass $\times 10^6$	static/kinetic, frozen winding (Docs. 286, 314)	localisation without disorder	**[S]**
π flux through triangle	winding number, $\mathbb{Z}_3$ phase ω^k (Docs. 314, 321)	Berry phase around loop	**[S]**

All four parallels are structurally established **[S]**. The physical question — whether the kagome system is a low-energy realisation of the FFGFT Galois structure — remains open and is not claimed here.

Open questions [S].

1. Is the Potts KT transition (separation of $\mathbb{Z}_3$-ordered / algebraically correlated) identifiable with the FFGFT confinement transition ($T_R = 0$ enforced / merely correlated) at finite temperature?
2. Pei et al. announce for two holes "polaron-mediated interactions [...] and possible binding, pairing". Two holes each with $T_R \neq 0$ could form a $T_R = 0$ state in FFGFT language. Is hole-pair binding on the kagome lattice a meson analogue?
3. The twelve sites of the self-trapping cluster versus $24 = 12 + 12$ from Doc. 314 — coincidence or structure?

Source. Pei, Y.; Chen, S. A.; Castelnovo, C.; Moessner, R.: *Kinetic Kagome Magnetism: From Self-Trapping RVB Polarons to Semiclassical Correlations*. Phys. Rev. Lett. **137**, 106702, published 31 August 2026. DOI: 10.1103/zbxv-vtq8. Open Access (Editors' Suggestion).